USEMO 2025 Problem 1

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What the sigma?

???

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§1 Problem Statement

Problem. Find all real numbers λ for which there exists an integer $n \geq 2$ and an arithmetic progression $(a_0, a_1, \ldots, a_n)$ of real numbers (in that order) such that the identity

$$(X - \lambda)(X - \lambda^2) \ldots (X - \lambda^n) = a_0 X^n + a_1 X^{n-1} + a_2 X^{n-2} + \cdots + a_n$$

is true for every real number X.

> **Remark.** I suggest all readers(if any) to try this problem before reading the rest of this document.

§2 Rough Work/Thought Process

Be warned, this is not going to be pretty, nor straightforward. Some calculations or simplifications may not be correct, as I'm just reading off whatever I did on my rough work.

§2.1 Any solutions here?

The first thing anyone should do is try out some easy cases. Let's start out with $n = 2$. Expanding, we get:

$$X^2 - \lambda^2 X - \lambda X + \lambda^3 = a_0 X^2 + a_1 X + a_2$$

Or, equivalently,

$$X^2 - (\lambda^2 + \lambda)X + \lambda^3 = a_0 X^2 + a_1 X + a_2$$

Since all a_i are in an arithmetic progression, it should follow that $a_2 - a_1 = a_1 - a_0$. Testing this gives:

$$\lambda^3 + \lambda^2 + \lambda = -\lambda^2 - \lambda - 1$$

And this leads to the only real solution being $\lambda = -1 \in \mathbb{R}$. We've found our first solution! You can try out some other cases on your own, and now it's pretty clear what our conjecture is: $\lambda = -1$ is the only real number that works.

§2.2 A very sad failed attempt

The next thing I moved on to was doing something with the arithmetic progression. Letting δ be the common difference, I wrote:

$$a_k = a_0 + k\delta$$

Next, I just let the given expression be a polynomial. Here's basically exactly the stuff I wrote down on my notebook:

$$\begin{aligned} P(X) &= (X - \lambda)(X - \lambda^2)\dots(X - \lambda^n) \\ &= a_0 X^n + (a_0 + \delta)X^{n-1} + \dots + (a_0 + n\delta) \\ &= a_0 X^n + a_0 X^{n-1} + \delta X^{n-1} + a_0 X^{n-2} + 2\delta X^{n-2} + \dots + a_0 + n\delta \\ &= a_0(X^n + \dots + 1) + \delta(X^{n-1} + 2X^{n-2} + \dots + n) \end{aligned}$$

Or, in other words:

$$P(X) = a_0 \sum_{k=0}^{n} X^k + \delta \sum_{i=0}^{n-1}((i+1)X^{n-1-i})$$

It was at this point though, that I realized this has absolutely nothing to do with the actual problem. This is just a dirty mess of symbols. So, let's take a step back.

§2.3 Vieta!?

The next thing I tried was the $n = 3$ case, and got nowhere with it because it was just far too cluttered for me to do anything with it. Well, what I tried after that turned out to be pretty useful:

$$\begin{aligned} P(X) &= (X - \lambda)(X - \lambda^2) \dots (X - \lambda^n) \\ &= a_0 X^n + (a_0 + \delta) X^{n-1} + (a_0 + 2\delta) X^{n-2} + \cdots + (a_0 + n\delta) \end{aligned}$$

While, again, this does just look like a hot mess of nothing, it led me to try out Vieta. So for $P(X) = 0$, observe that we have:

$$\lambda + \lambda^2 + \cdots + \lambda^n = -\frac{a_0 + \delta}{a_0}$$

Or, in other words:

$$\frac{\lambda(\lambda^{n-1} - 1)}{\lambda - 1} = -\frac{a_0 + \delta}{a_0}$$

Seems like we're getting somewhere. But that a_0 in the denominator is quite annoying, if only it could be equal to 1...

§2.4 A funny coincidence

I mentioned how the expression would be a lot nicer if a_0 was just equal to 1. Well, even though it actually just is(try expanding the original $P(X)$), I didn't really see it at first. So it just led me to try and play around with $P(X)$, giving birth to this:

$$Q(X) = (X - 1)P(X)$$

For some reason, I thought the extra 1 would somehow get rid of a_0 while expanding it, and leave me with just 1 as the coefficient of X^n. It's good that I thought this however, as it turned out to be quite interesting. Expanding this, we obtain:

$$\begin{aligned} Q(X) &= (X - 1)(a_0 X^n + a_1 X^{n-1} + \cdots + a_n) \\ &= (a_0 X^{n+1} + a_1 X^n + \cdots + a_n X) - (a_0 X^n - a_1 X^{n-1} - \cdots - a_n) \end{aligned}$$

Okay, let's simplify it further:

$$Q(X) = a_0 X^{n+1} + (a_1 - a_0) X^n + (a_2 - a_1) X^{n-1} + \cdots + (a_n - a_{n-1}) X - a_n$$

Remember that a_0 is forced to be 1 though, and $a_{i+1} - a_i = \delta$. Thus, we have:

$$\boxed{Q(X) = X^{n+1} + \delta(X^n + \cdots + X) - a_n}$$

We have hope!

§2.5 Welcome back, Vieta!

We noticed earlier that we just wanted $a_0 = 1$. In the process of doing so, however, we obtained a much more beautiful polynomial. Therefore, let's just consider the following:

$$\begin{aligned} Q(X) &= (X-1)(X-\lambda)\dots(X-\lambda^n) \\ &= X^{n+1} + \delta(X^n + \cdots + X) - a_n \end{aligned}$$

We can peacefully apply Vieta's on Q that gives us:

$$1 + \lambda + \lambda^2 + \cdots + \lambda^n = -\delta$$

Which gives us that:

$$\delta = -\left(\frac{\lambda^{n+1} - 1}{\lambda - 1}\right)$$

§2.6 The final stretch

We now have the coefficients of X^n through X in Q. This, however, was obtained directly through Vieta. There's another way to find the coefficients – yes, obviously by expanding the original product. A few small cases would lead you to the conclusion that determining the coefficient of X this way is the best and simplest way to do so, it doesn't matter obviously as the coefficients of X^n through X are the same. So let's look at a few cases:

$$(X-1)(X-\lambda)(X-\lambda^2)$$

Expanding this gives us that the coefficient of X is equal to $(\lambda + \lambda^2 + \lambda^3)$. Okay, what about this then?

$$(X-1)(X-\lambda)(X-\lambda^2)(X-\lambda^3)$$

Brutally expanding this gives us that the coefficient of X equals $-(\lambda^3 + \lambda^4 + \lambda^5 + \lambda^6)$. Maybe you've already seen the pattern, but I'll give one more example:

$$(X-1)(X-\lambda)(X-\lambda^2)(X-\lambda^3)(X-\lambda^4)$$

Here, one can check that the coefficient of X is $(\lambda^6 + \lambda^7 + \lambda^8 + \lambda^9 + \lambda^{10})$. Let's write all of these results down for clarity:

$$\begin{aligned} \lambda^2 &: && \lambda + \lambda^2 + \lambda^3 \\ \lambda^3 &: && -(\lambda^3 + \lambda^4 + \lambda^5 + \lambda^6) \\ \lambda^4 &: && \lambda^6 + \lambda^7 + \lambda^8 + \lambda^9 + \lambda^{10} \end{aligned}$$

Where the left side of λ^i represents the product $(X-1)(X-\lambda)\dots(X-\lambda^i)$, and the right side represents the coefficient of X of the polynomial in the left side.

Interesting. I'm sure the pattern is noticable now – if not, try and ponder upon it for a bit. If you do have a guess, I'll write down the case for λ^5, and you can check your answer.

$$\lambda^5 : \qquad -(\lambda^{10} + \lambda^{11} + \lambda^{12} + \lambda^{13} + \lambda^{14} + \lambda^{15})$$

I'll count this as a thinking exercise(definitely not because I'm too lazy to write up my thought process on how I got this answer), but the general form is:

$$\lambda^k : \qquad (-1)^k \lambda^{\frac{k(k-1)}{2}} \left(\sum_{i=0}^{k} \lambda^i \right)$$

Yes, it looks scary, but it isn't that hard to understand once you get where it actually comes from. Just for simplification purposes, let's rewrite this as:

$$(-1)^k \lambda^{\frac{k(k-1)}{2}} \left(\frac{\lambda^{k+1} - 1}{\lambda - 1} \right)$$

§2.7 We're done!

Aha! So now we have the coefficient of X expressed in two ways. Obviously, we equate them:

$$(-1)^n \lambda^{\frac{n(n-1)}{2}} \left(\frac{\lambda^{n+1} - 1}{\lambda - 1} \right) = - \left(\frac{\lambda^{n+1} - 1}{\lambda - 1} \right)$$

$$\iff \left(\frac{\lambda^{n+1} - 1}{\lambda - 1} \right) \left[(-1)^n \lambda^{\frac{n(n-1)}{2}} + 1 \right] = 0$$

So now we have two cases. First, consider when:

$$\frac{\lambda^{n+1} - 1}{\lambda - 1} = 0$$

This only occurs when $|\lambda| = 1$, but here $\lambda \neq 1$(and it can easily be checked that it doesn't work anyway), so this is just reduced to our original verified solution of $\lambda = -1$ and $n = 2$. Next,

$$\left[(-1)^n \lambda^{\frac{n(n-1)}{2}} + 1 \right] = 0$$

$$\iff \lambda^{\frac{n(n-1)}{2}} = (-1)^{1-n}$$

$$\iff |\lambda^{\frac{n(n-1)}{2}}| = 1$$

Since $\frac{n(n-1)}{2} > 0$, this forces $|\lambda| = 1$, which is just the other case again. We're done!